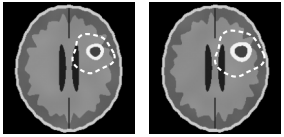
A At each follow up MRI after treatment, has the tumour progressed?

PRISM keeps image and clinical-history evidence apart and reports them side by side

1 Inputs

MRI, released masks, record



previous current

T1-contrast and FLAIR MRI

surgery scan

chemoRT

age · sex · IDH · MGMT

treatment dates, clinical record

2 Measure

27 measurements per scan

Released masks

6 volumes, 13 shape and intensity, 8 change enhancing 4.0 → 6.0 cc

Phase of care

before RT end, early (≤ 12 weeks), later here: early, 8 weeks

3 Score

3 ridge logistic models

Imaging 0.64

Combined 0.33

Clinical history 0.21

synthetic example values

4 Confidence

conformal, median rule, lookup

80% conformal set

{progression, no progression}

both outcomes

Model agreement

disagree

imaging above, clinical history below their cohort medians

Reliability card

few cases

1 of 9 such scans was progression

5 Report

Qwen3.8-27B on vLLM, local

Record for the scan

every value with its source

Local language model

one GPU on site; sees no images, does no arithmetic

Deterministic checker

every number and flag must be in the record, or the text is rejected

rejected: no report

PRISM surveillance report SYNTHETIC EXAMPLE

Imaging		0.64
Combined		0.33
Clinical history		0.21

Enhancing tumour is 2.0 cc larger than at the previous scan. The two models disagree, and the 80% conformal set holds both outcomes. This situation has few cases (1 of 9 scans was progression).

B How it was evaluated

MU-Glioma-Post

public development cohort, 5 × 5 cross validation by patient

Models locked

fitted once on MU; plans committed

LUMIERE, UCSF

applied unchanged

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reports checked

Python 3.11 · scikit-learn 1.7.2 · vLLM 0.31.0 · one NVIDIA H100